

School of Computer Science and Artificial Intelligence

Lab Assignment # 3.2

Program	: B. Tech (CSE)
Specialization	: –
Course Title	: AI Assisted coding
Course Code	:
Semester	: II
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Task 1:

- Progressive Prompting for Calculator Design: Ask the AI to design a simple calculator program by initially providing only the function name. Gradually enhance the prompt by adding comments and usage examples.

Prompt: Create a simple calculator that takes two numbers and an operator as input and performs the corresponding operation

```
def calculator():  
  
    a = float(input("Enter first number: "))  
    b = float(input("Enter second number: "))  
    op = input("Enter operator (+, -, *, /): ")  
  
    if op == "+":  
        result = a + b  
    elif op == "-":  
        result = a - b  
    elif op == "*":  
        result = a * b  
    elif op == "/":  
        if b != 0:  
            result = a / b  
        else:  
            print("Error: Division by zero")  
            return  
    else:  
        print("Invalid operator")  
        return  
  
    print("Result:", result)  
  
calculator()
```

```
C:\Users\aelur\OneDrive\Desktop\Ai_assistant>C:/Python314
Enter first number: 10
Enter second number: 20
Enter operator (+, -, *, /): +
Result: 30.0
```

Explanation:

- This code defines a simple calculator function that prompts the user for two numbers and an operator.
- It then performs the corresponding arithmetic operation based on the operator provided and displays the result.
- The code also handles division by zero and invalid operators gracefully.

Task 2 :

Refining Prompts for Sorting Logic: Start with a vague prompt for sorting student marks, then refine it to clearly specify sorting order and constraints.

Prompt : Write a program to sort student marks

```
def sort_student_marks(marks):

    valid_marks = []

    for mark in marks:
        if 0 <= mark <= 100:
            valid_marks.append(mark)
    sorted_marks = sorted(valid_marks, reverse=True)

    return sorted_marks

marks = [78, 45, 120, -5, 89, 66, 92]

print("Original Marks:", marks)
print("Sorted Marks (Descending Order):", sort_student_marks(marks))
```

```
C:\Users\aelur\OneDrive\Desktop\Ai_assistant>C:/Python314
Original Marks: [78, 45, 120, -5, 89, 66, 92]
Sorted Marks (Descending Order): [92, 89, 78, 66, 45]
```

Explanation:

- The function `sort\_student\_marks` filters out invalid marks (outside 0-100 range) and returns a list of valid marks sorted in descending order.
- The original list is [78, 45, 120, -5, 89, 66, 92], and after filtering and sorting, the result is [92, 89, 78, 66, 45].

Task 3:

Few-Shot Prompting for Prime Number Validation: Provide multiple input-output examples for a function that checks whether a number is prime. Observe how few-shot prompting improves correctness.

Prompt: Write a function to check whether a number is prime.

```
def is_prime(n):  
    if n <= 1:  
        return False  
  
    for i in range(2, int(n ** 0.5) + 1):  
        if n % i == 0:  
            return False  
  
    return True  
  
num = int(input("Enter a number: "))  
  
if is_prime(num):  
    print(num, "is a Prime Number")  
else:  
    print(num, "is Not a Prime Number")
```

```
C:\Users\aelur\OneDrive\Desktop\Ai_assistant>C:/Python314  
Enter a number: 10  
10 is Not a Prime Number
```

Explanation:

- The function `is\_prime` checks if a number is prime by testing divisibility up to the square root of the number.
- If any divisor is found, it returns False; otherwise, it returns True.
- The main program prompts the user for a number and uses `is\_prime` to determine and display whether it is prime.

Task 4:

Prompt-Guided UI Design for Student Grading System: Create a user interface for a student grading system that calculates total marks, percentage, and grade based on user input.

Prompt: Guided UI Design for Student Grading System

```

<!DOCTYPE html>
<html>
<body>

<h3>Student Grading System</h3>

<input id="m1" placeholder="Subject 1">
<input id="m2" placeholder="Subject 2">
<input id="m3" placeholder="Subject 3">
<input id="m4" placeholder="Subject 4">
<input id="m5" placeholder="Subject 5">

<button onclick="calc()">Calculate</button>

<p id="out"></p>

<script>
function calc() {
    let total =
        Number(m1.value) + Number(m2.value) +
        Number(m3.value) + Number(m4.value) +
        Number(m5.value);

    let per = (total / 500) * 100;
    let grade = per >= 75 ? "A" : per >= 60 ? "B" : per >= 40 ? "C" : "Fail";

    out.innerHTML =
        "Total: " + total + "<br>" +
        "Percentage: " + per.toFixed(2) + "%<br>" +
        "Grade: " + grade;
}
</script>

</body>
</html>

```

Output:

```

Subject 1 = 80
Subject 2 = 75
Subject 3 = 70
Subject 4 = 85
Subject 5 = 90

```

```

Total: 400
Percentage: 80.00%
Grade: A

```

Explanation :

- The user enters marks for five subjects and clicks the Calculate button.
- The program computes the total marks, percentage, and assigns a grade based on the percentage.
- The result is displayed clearly on the screen below the button.

Task 5 :

Analyzing Prompt Specificity in Unit Conversion Functions: Improving a Unit Conversion Function (Kilometers to Miles and Miles to Kilometers) Using Clear Instructions.

Prompt: "Write a unit conversion function."

```
def convert_distance():  
    print("Unit Conversion Menu")  
    print("1. Kilometers to Miles")  
    print("2. Miles to Kilometers")  
  
    choice = int(input("Enter your choice (1 or 2): "))  
    value = float(input("Enter the value to convert: "))  
  
    if choice == 1:  
        miles = value * 0.621371  
        print("Miles:", round(miles, 3))  
    elif choice == 2:  
        kilometers = value / 0.621371  
        print("Kilometers:", round(kilometers, 3))  
    else:  
        print("Invalid choice")  
  
convert_distance()
```

```
C:\Users\aelur\OneDrive\Desktop\Ai_assistanr>C:/Python  
Unit Conversion Menu  
1. Kilometers to Miles  
2. Miles to Kilometers  
Enter your choice (1 or 2):
```

Explanation :

- The `convert\_distance` function provides a simple interface for converting between kilometers and miles.
- It prompts the user for a conversion choice and a value, then performs the appropriate conversion.
- The result is displayed with three decimal places for precision.